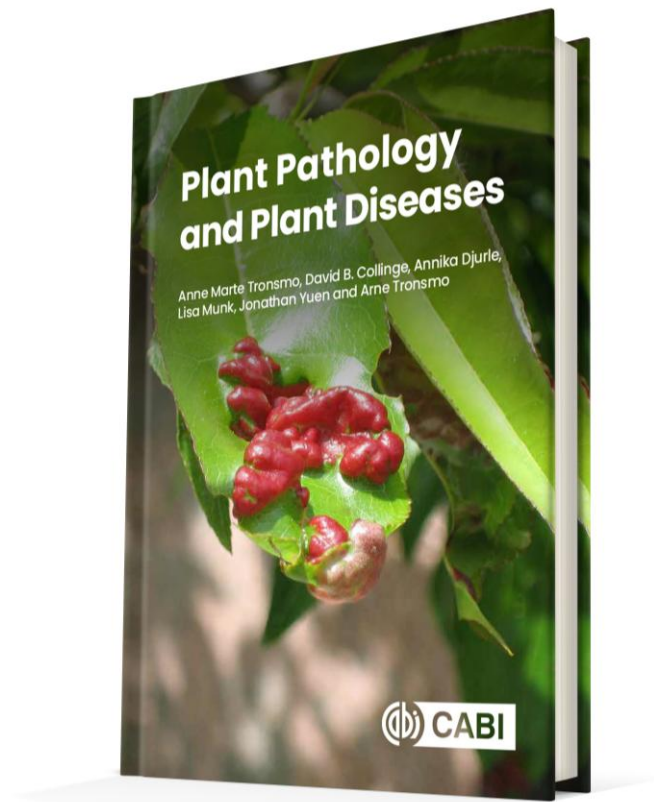




Plant Pathology and Plant Diseases

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Fig. 11.1 Schematic model for of active defence. (© D.B.Collinge.)

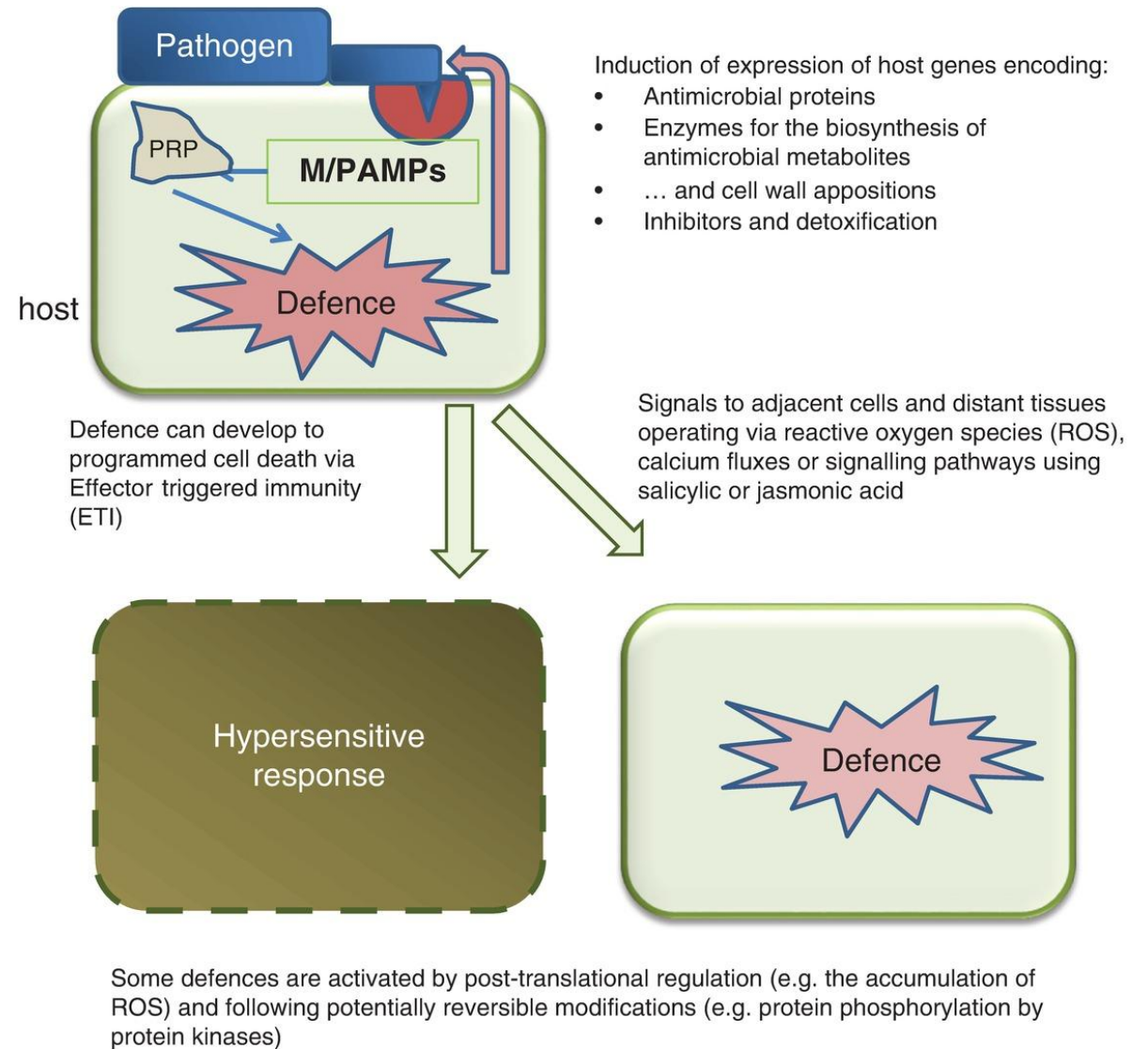


Fig. 11.2.A-D. Papillae (cell wall appositions) with different histochemical stains.

Epidermis of barley (left) and wheat (right) both inoculated with the barley powdery mildew fungus, *Blumeria graminis* f.sp. *hordei* (24 hours after inoculation). (A) and (B) stained with lacmoid, indicating callose.

(From Wei *et al.*, 1994. *PMPP* 45. 467-484.)

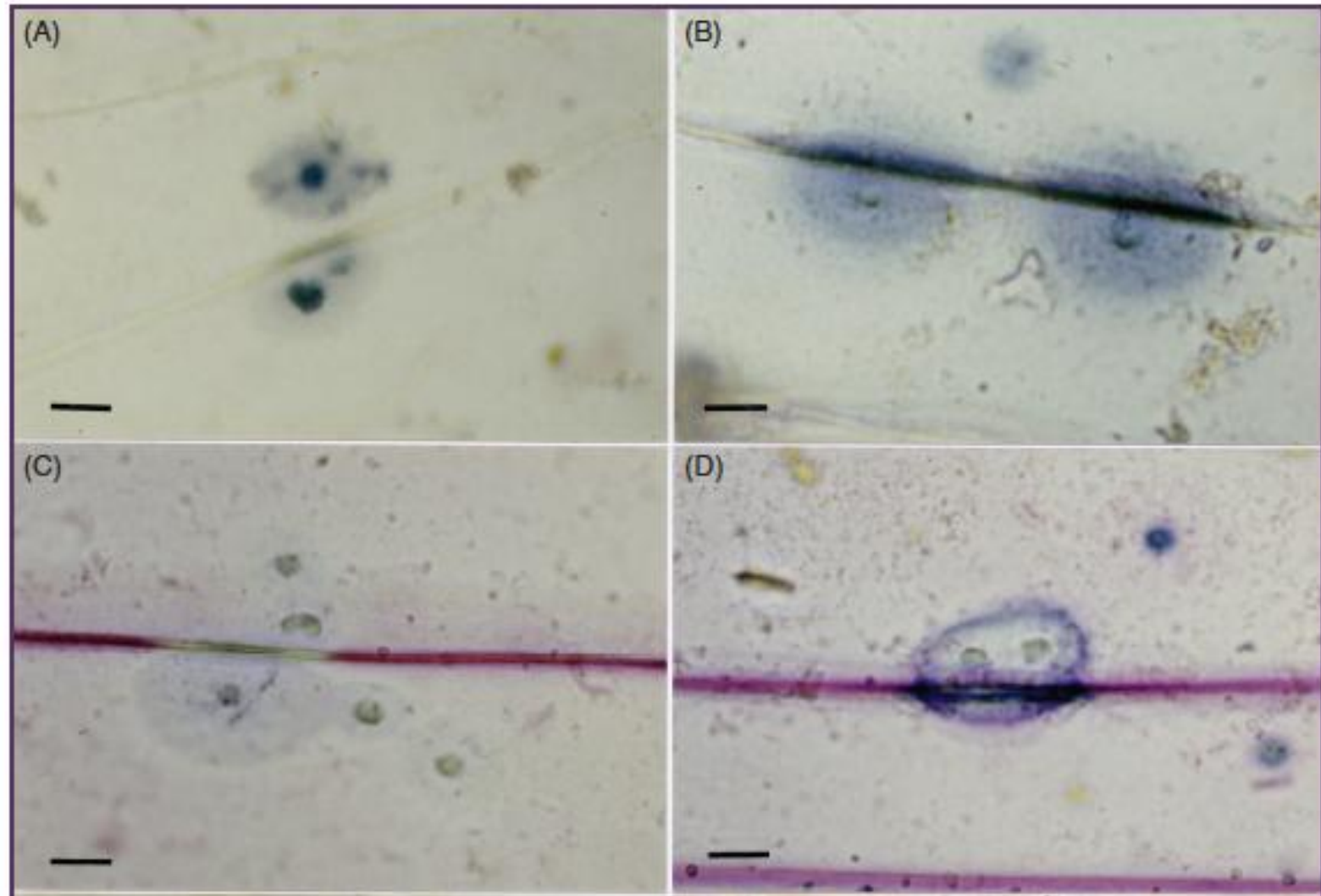


Fig. 11.2. E-H. Papillae (cell wall appositions) with different histochemical stains. Epidermis of barley (left) and wheat (right) both inoculated with the barley powdery mildew fungus, *Blumeria graminis* f.sp. *hordei* (24 hours after inoculation). E and F Phloroglucinol-HCl, indicating lignin. G and H. Sakaguchi staining method, indicating guanidine compounds. Bar= 20 µm. (From Wei *et al.*, 1994. *PMPP* 45. 467-484.)

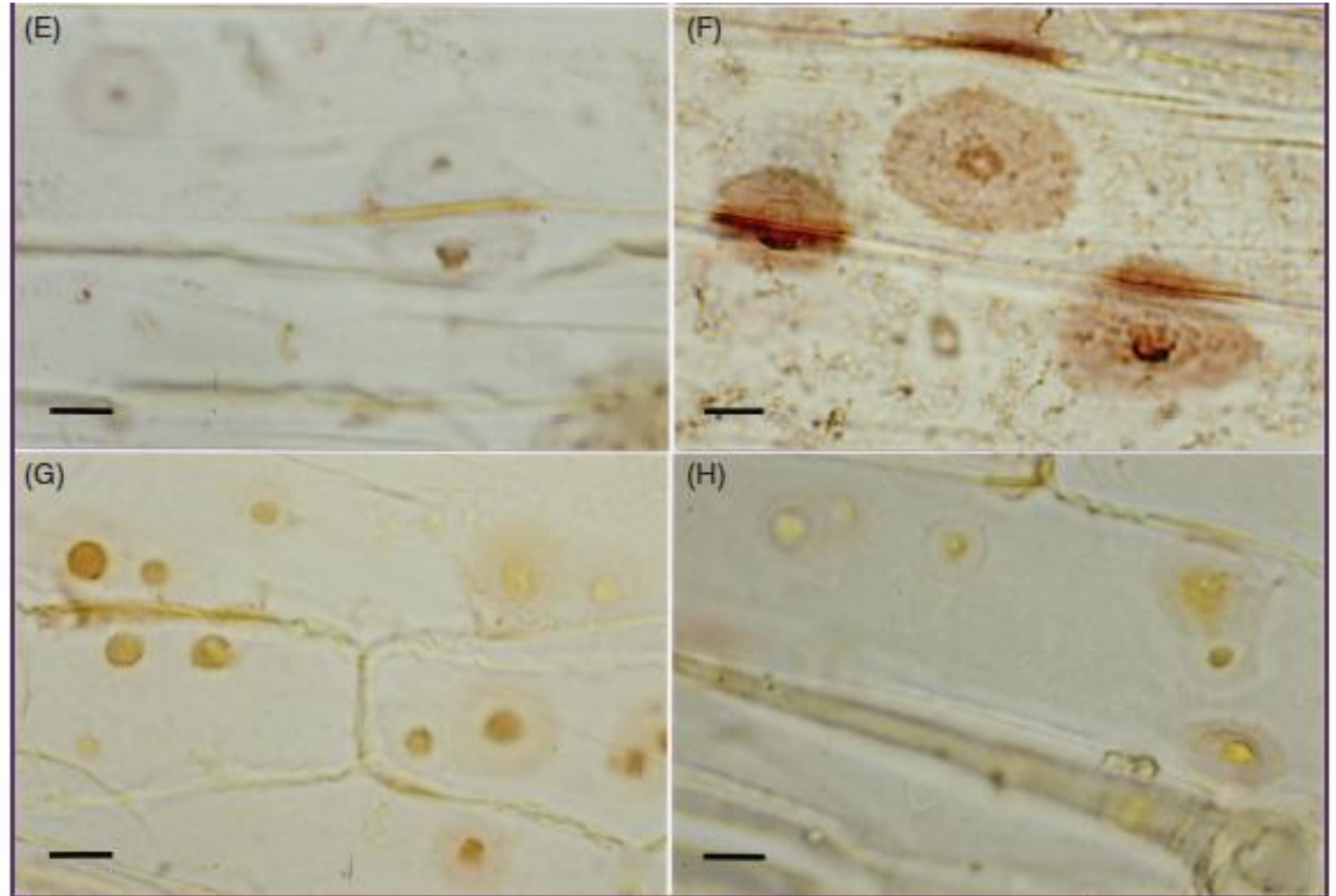


Fig. 11.3. DAB staining for papillae and the hypersensitive response (HR). Detection of reactive oxygen species using hydrogen peroxide in barley inoculated with the barley powdery mildew fungus, *Blumeria graminis* f.sp. *hordei* in papillae (A) 6 and (B) 24 hours after inoculation and (C) in cells exhibiting an HR 24 hours after inoculation.

Diaminobenzidine (1 mg ml⁻¹) staining for 8 hours. P, primary germ tube; a, appresorial germ tube; pa, papilla; HR, hypersensitive response.

(From Thordal-Christensen *et al.*, 1997. *TPJ* 11. 1187-1194)

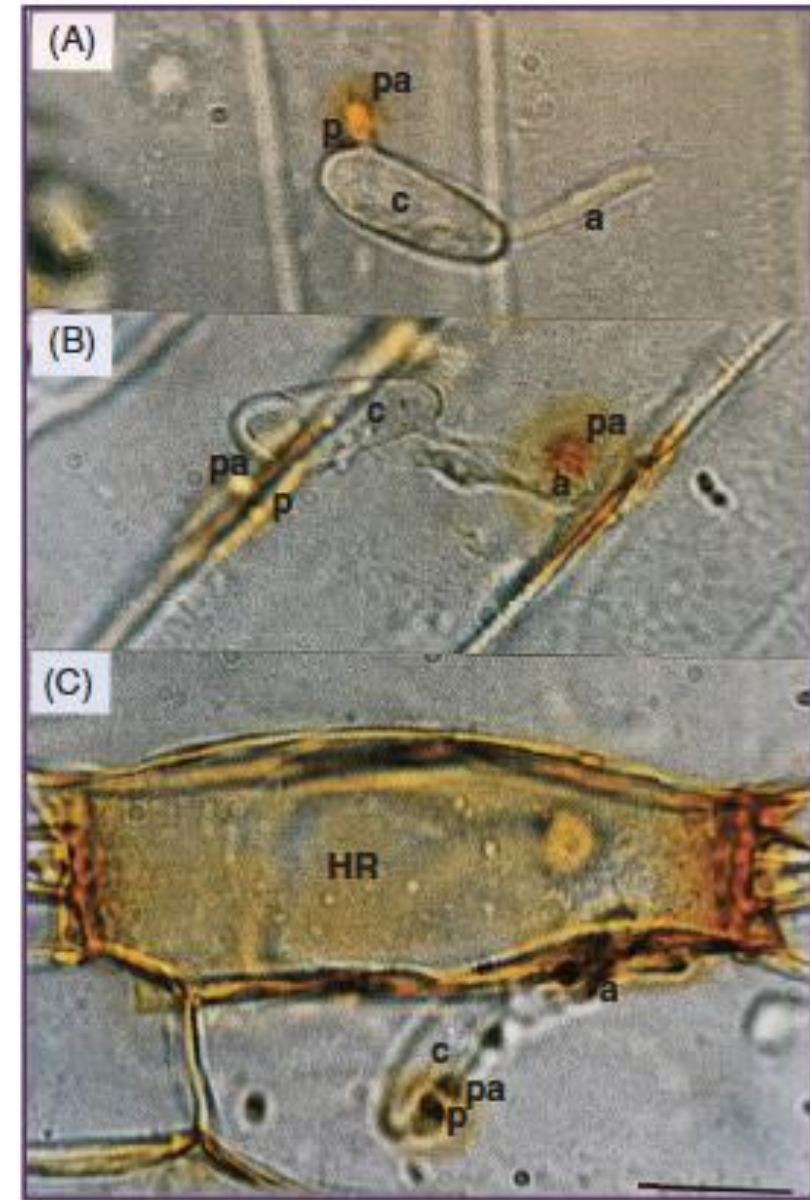
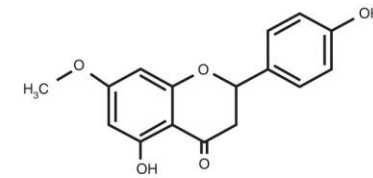
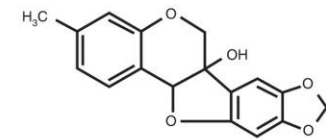


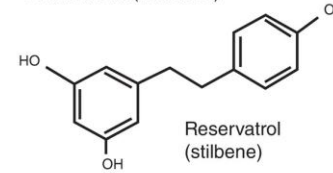
Fig. 11.4 Some structures of phytoalexins and phytoanticipins. (© D.B. Collinge.)



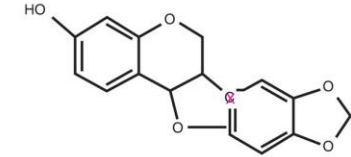
Sakuranetin (flavonoid)



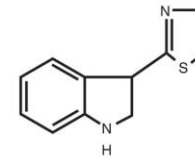
Pisatin (isoflavonoid)



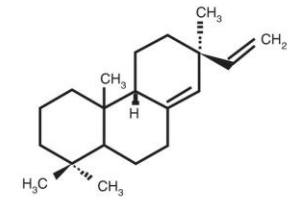
Resveratrol
(stilbene)



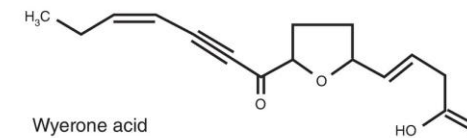
Maackiain (isoflavonoid)



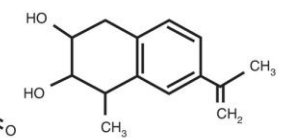
Camalexin
(indole)



Momilactone B
(diterpene)

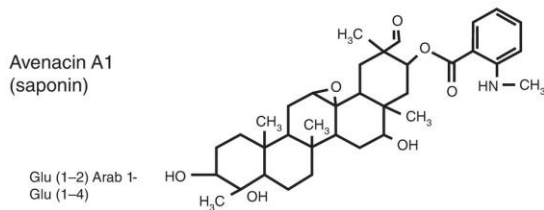


Wyerone acid



Rishitin (sesquiterpene)

Avenacin A1
(saponin)



Glu (1-2) Arab 1-
Glu (1-4)